Also, if we differentiate $\frac{\partial F}{\partial y}(x,y)$ partially with respect to y we obtain $\frac{\partial^2 F}{\partial y^2}$; which is called the second order partial derivatives of F with respect to y. So for any function F defined on any subset $\{(x,y) \mid a < x < b, c < y < d\} \subset \mathbb{R}^2$ we have the following notation :

$$\frac{\partial^2 F}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial F}{\partial x}\right) = F_{xx}, \quad \frac{\partial^2 F}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial F}{\partial y}\right) = F_{xy}$$

$$\frac{\partial^2 F}{\partial y \partial x} = \frac{\partial}{\partial y}\left(\frac{\partial F}{\partial x}\right) = F_{yx}, \quad \frac{\partial^2 F}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial F}{\partial y}\right) = F_{yy}$$

All the above are called second order partial derivatives of F. Similarly we can define higher order partial derivatives. For example, $\frac{\partial^3 F}{\partial y^2 \partial x} = \frac{\partial}{\partial y}\left(\frac{\partial}{\partial y}\left(\frac{\partial F}{\partial x}\right)\right)$, and $\frac{\partial^3 F}{\partial x \partial y \partial x} = \frac{\partial}{\partial x}\left(\frac{\partial}{\partial y}\left(\frac{\partial F}{\partial x}\right)\right)$.

Next we shall see more examples on partial differentiation.

Example 8.13

Let $f(x,y) = \sin(xy^2) + e^{x^3+5y}$ for all $(x,y) \in \mathbb{R}^2$. Calculate $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial^2 f}{\partial y \partial x}$ and $\frac{\partial^2 f}{\partial x \partial y}$.

Solution

First we shall calculate $\frac{\partial f}{\partial x}(x,y)$. Note that f is a sum of two functions and so

$$\begin{aligned}\frac{\partial f}{\partial x} &= \frac{\partial}{\partial x}\sin(xy^2) + \frac{\partial}{\partial x}\left(e^{x^3+5y}\right)\\ &= \cos(xy^2)\frac{\partial}{\partial x}(xy^2) + e^{x^3+5y}\frac{\partial}{\partial x}(x^3+5y)\\ &= \cos(xy^2)y^2 + e^{x^3+5y}3x^2.\end{aligned}$$

Similarly,

$$\begin{aligned}\frac{\partial f}{\partial y} &= \frac{\partial}{\partial y}\sin(xy^2) + \frac{\partial}{\partial y}\left(e^{x^3+5y}\right)\\ &= \cos(xy^2)\frac{\partial}{\partial y}(xy^2) + e^{x^3+5y}\frac{\partial}{\partial y}(x^3+5y)\\ &= \cos(xy^2)2xy + 5e^{x^3+5y}.\end{aligned}$$

Next we consider,

$$\begin{aligned}\frac{\partial^2 f}{\partial y \partial x} &= \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial}{\partial y}\left(y^2\cos(xy^2) + 3x^2e^{x^3+5y}\right)\\ &= \frac{\partial}{\partial y}(y^2\cos(xy^2)) + \frac{\partial}{\partial y}\left(3x^2e^{x^3+5y}\right)\\ &= 2y\cos(xy^2) + y^2\left(-\sin(xy^2)2xy\right) + 3x^2e^{x^3+5y}5\\ &= 2y\cos(xy^2) - 2xy^3\sin(xy^2) + 15x^2e^{x^3+5y}.\end{aligned}$$

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